

Question 1: Choose the correct answers:-
 (الاجابات في نص ورقة الأسئلة)

- Classes are useful because they
 - are removed from memory when not in use.
 - bring together all aspects of an entity in one place.
 - permit data to be hidden from other classes.
 - can closely model objects in the real world.
- A friend function can be used to
 - allow access to classes whose source code is unavailable.
 - allow access to an unrelated class.
 - mediate arguments between classes.
 - increase the versatility of an overloaded operator.
- Inheritance is a way to
 - add features to existing classes without rewriting them.
 - pass arguments to objects of classes.
 - make general classes into more specific classes.
 - improve data hiding and encapsulation.
- In a class definition, data or functions designated private are accessible
 - to any function in the program.
 - to member functions of that class.
 - only if you know the password.
 - only to public members of the class.
- Virtual functions allow you to
 - use the same function call to execute member functions of objects from different classes.
 - create an array of type pointer-to-base class that can hold pointers to derived classes.
 - group objects of different classes so they can all be accessed by the same function code.
 - create functions that can never be accessed.
- In any object of the class of which it is a member. The scope-resolution operator usually
 - resolves ambiguities.
 - tells what base class a class is derived from.
 - specifies a particular class.
 - limits the visibility of variables to a certain function.
- An abstract class is useful when
 - no classes should be derived from it.
 - no objects should be instantiated from it.
 - there are multiple paths from one derived class to another.
 - you want to defer the declaration of the class.
- Assume a class Derv that is privately derived from class Base. An object of class Derv located in main() can access
 - protected members of Derv.
 - private members of Derv.
 - public members of Derv.
 - public members of Base.
- A member function can always access the data.
 - in the object of which it is a member.
 - in the class of which it is a member.
 - in the public part of its class.
 - in any object of the class of which it is a member.

10. A pure virtual function is a virtual function that
causes its class to be abstract.
is used in a base class.

- b. returns nothing
- d. takes no arguments

Question 2: Illustrate the following:

- (a) Life cycle of an object in object-oriented programming
- (b) Illustrate the types of inheritance with the accessibilities

Question 3: Write code syntax.

(a) Levels of inheritance

(b) Overriding member function

Question 5: Write an OOP Program:

Define a class name Address as follows:

- 1) Data members are houseNo and streetName, cityName, stateName, and zipcode.
- 2) Define a parameter constructor and a destructor.
- 3) Define an accessor member function to print the address.